Tushika Garg

+91-7078323902 | gargtushikabsr@gmail.com | linkedin.com/in/tushika-garg | github.com/Tushika2024| Leetcode

Computer Science undergraduate with strong fundamentals in Data Structures and Algorithms. Proficient in Python, C++, and SQL, with experience in building and deploying machine learning systems. Currently working with production data and backend systems as a Software Developer Intern.

Education

Thapar Institute of Engineering and Technology — *Patiala, Punjab*
B.E. in Computer Science and Business Systems; CGPA: ***9.92/10.0*** — *Aug 2024 – May 2028*

Renaissance School — *Bulandshahr, Uttar Pradesh*
Senior Secondary Examination CBSE; Percentage: ***95.6%*** — *Apr 2022 – Mar 2023*

Work Experience

Software Developer Intern — *Jun 2026 – Present*
Kanha Milk Testing Equipments Pvt. Ltd. — *Bulandshahr, UP*

- **Migrated 10,000+ legacy Excel records** into a structured MySQL database via a Python ETL pipeline, enforcing data standardization and reducing query overhead across distributed production sources.
- Developing a FastAPI based complaints portal with automated daily complaint analytics dashboard, enabling operational stakeholders to monitor trends and support data-driven resolution decisions.

Projects

CreditShield | ***Python, Sklearn, Optuna, SHAP, Streamlit*** | GitHub | Live — *Apr 2026 – Jun 2026*

- Engineered a **credit underwriting pipeline** on 30K+ records using Optuna-tuned XGBoost with **ROC-AUC 0.82**, reject inference, and **15+ behavioral risk features** to support data-driven credit decisioning.
- Designed a risk-segmented credit limit engine with SHAP explainability and configurable policy guardrails, delivering auditable, compliance-ready customer credit decisions via Streamlit.

SentinelLabel | ***Python, Transformers, NLP, Sklearn, Cross-Encoders*** | GitHub — *Feb 2026 – Apr 2026*

- Developed a semantic label auditing framework across a 14K+ NLP corpus, flagging **268 mislabeled samples** with **82% precision**, achieving **3.6× improvement** over random sampling.
- Applied UMAP dimensionality reduction and HDBSCAN clustering to surface structural data inconsistencies, enforcing governance standards for downstream product and model decisions.

Sleep Behavior Modeling for T2DM | ***Python, Sklearn, Pandas, NumPy*** | GitHub — *Nov 2025 – Jan 2026*

- Conducted EDA and statistical modeling on the AI-READI wearable health dataset, converting raw sensor data into structured, decision-ready features linked to Type-2 Diabetes risk.
- Achieved **ROC-AUC 0.75** and **F1-score 0.84** on T2DM risk classification; manuscript on Pattern Recognition in Wearable Sleep Data currently under conference review.

Technical Skills

Programming Languages: C++, Python, C, SQL, HTML, CSS
Machine Learning: Supervised & Unsupervised Learning, Model Training, Model Evaluation
Deep Learning & NLP: Neural Networks, Transformers, Text Classification, Embeddings
Libraries & Frameworks: Scikit-learn, Pandas, NumPy, Streamlit, FastAPI, Hugging Face
Tools: Git, GitHub, VS Code, MySQL
Coursework: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering

Achievements

- **Merit I Scholarship:** Awarded for achieving Rank 1 in Branch (First Year), Thapar Institute, 2024–25.
- **Flipkart Gridlock 2.0 (2026):** Top 1000 of 10,600+ teams, ML-based urban traffic congestion challenge.
- **GirlScript Summer of Code (2025):** Open Source Contributor for real-world projects via GitHub.
- **HackTU 6.0 (2025):** Led frontend development and ideation during a 24-hour rapid prototyping event.
- **Best Student of the Year:** Renaissance School, Session 2022–2023, for all-round excellence.
- **Competitive Programming:** Solved 250+ DSA problems on LeetCode, improving code optimization and algorithmic efficiency.

Leadership & Extra-Curricular

- **Institution of Engineers (IEI):** Technical Department Member.
- **Student Council:** Batch Representative, demonstrating leadership, coordination, and decision-making in student affairs.
- **Design Thinking (Course Project):** Conducted empathy interviews with 20+ students in a 4-member Design Thinking team; defined problem statements around classroom engagement and communication barriers.